SimMonitor User Manual

SimMonitor is a Windows program that shows on your desktop how far your simulations have progressed. When a simulation finishes or stops, it sends an alert to your phone via Telegram.



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1. What SimMonitor does

- **Desktop window:** Each running simulation is shown as one line in a small window in the top-right corner of the screen. Supported programs are Fluent, FLOW-3D, Abaqus, Ansys Mechanical, OpenFOAM and TC-Python.
- **Information for each simulation:** number of CPU cores used, progress, time remaining and expected finish time.
- **Telegram notifications (optional):** A status report is sent every hour on the hour. You are told right away when a simulation finishes, stops or ends with an error.
- **No effect on your simulations:** SimMonitor only reads simulation files. It never stops or slows down the simulation programs.

2. What you need

Item	Details
Windows 10 or 11 (64-bit)	Install SimMonitor on the PC that runs the simulations
Internet connection	Used to download Rainmeter during setup and to send Telegram notifications
Telegram app	Needed only if you want notifications. It can be on your phone or your PC

You do not need administrator rights. However, if **Rainmeter**, the free program that draws the desktop window, is not installed, setup installs it too. At that point Windows asks you once whether to allow it.

3. Installing

3-1. Download the installer

Download `SimMonitorSetup-version.exe` from the Releases page.

3-2. Run the installer

Double-click the downloaded file. Administrator rights are not needed, so do not use [Run as administrator].

If a "Windows protected your PC" window appears, do the following. This window appears because SimMonitor was made by an individual and is not signed with a Microsoft certificate.

1. Click [More info] (the small text under the message).
2. Check that the app name is `SimMonitorSetup-version.exe`, then click [Run anyway].

(Optional) Check that the file has not been changed: You can compare it with the SHA-256 value shown on the Releases page. Run the command below in PowerShell and check that the value it shows is the same.

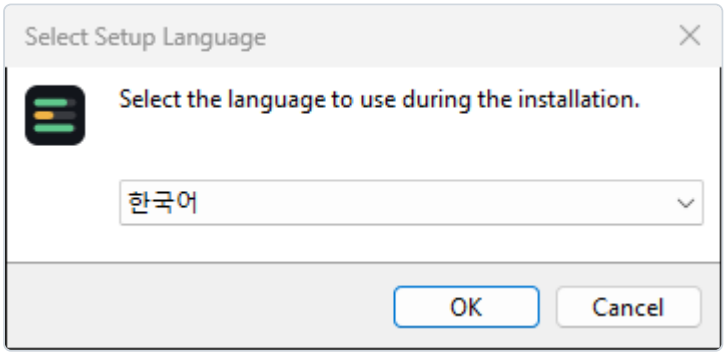
```
Get-FileHash "$env:USERPROFILE\Downloads\SimMonitorSetup-1.1.0.exe"
```

On a PC with **Smart App Control** turned on, setup is blocked instead of showing this window. See 11. Troubleshooting.

3-3. Setup wizard

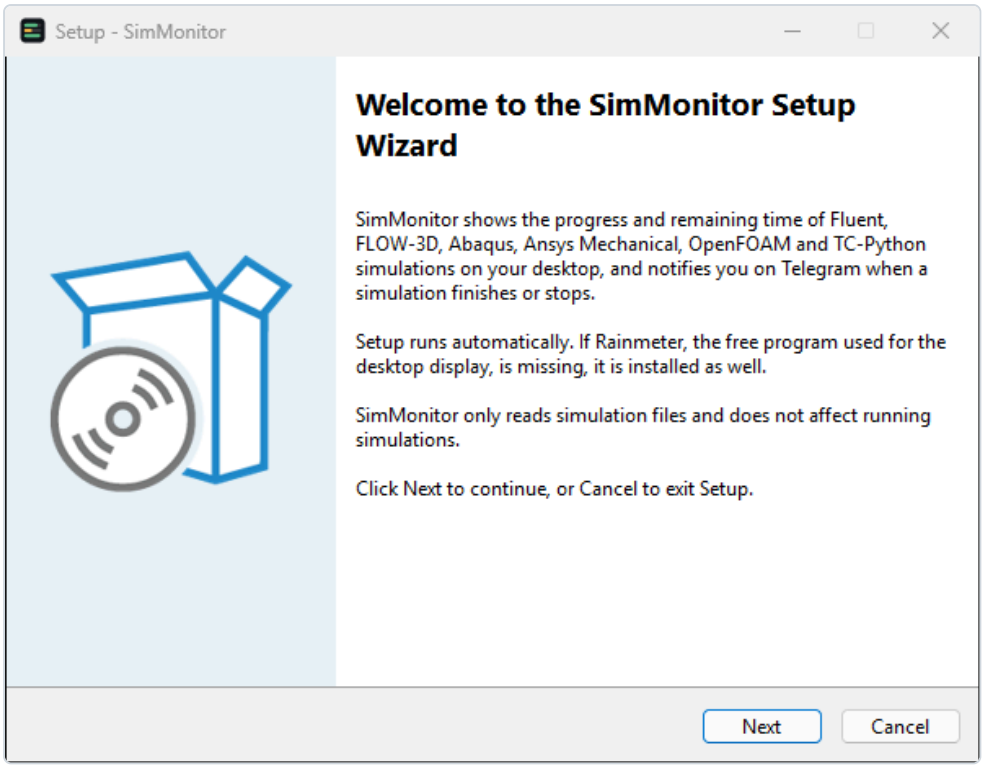
① Select Setup Language

Choose the setup language and click [OK]. SimMonitor uses the same language. You can change it later in the settings window.



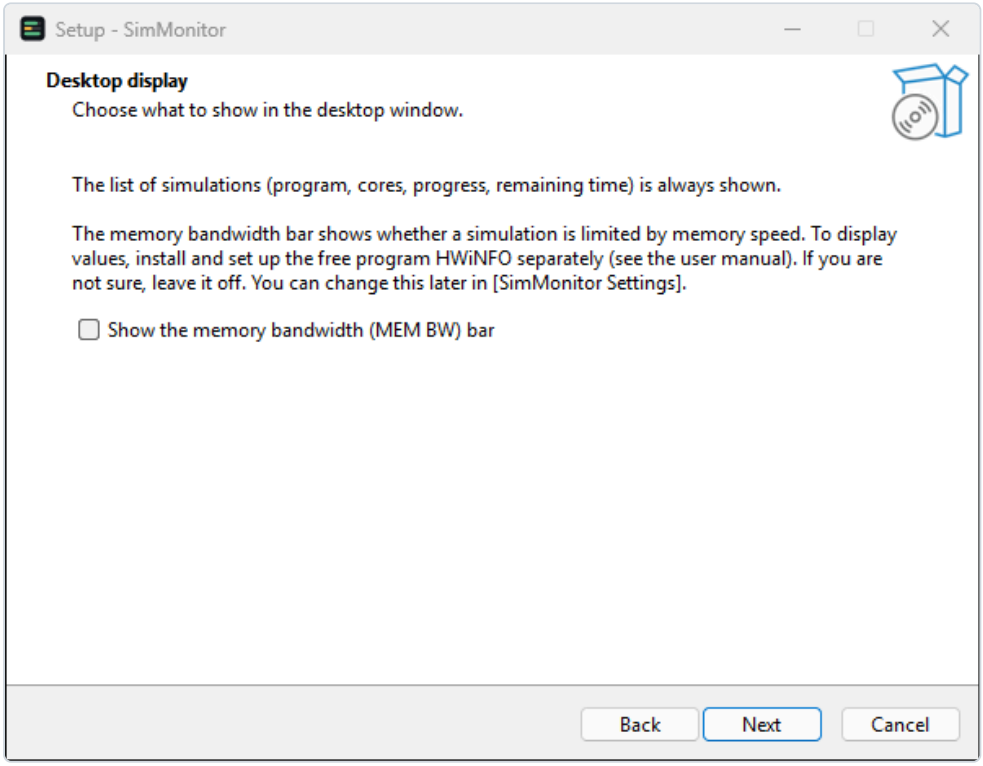
② Welcome screen

Click [Next].



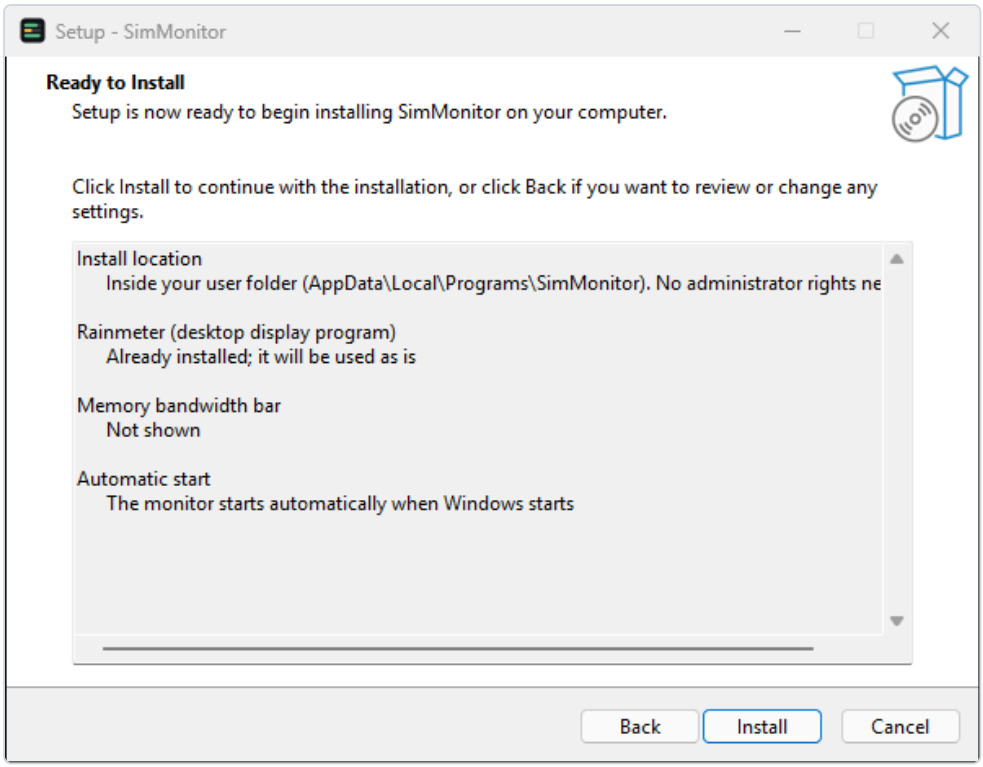
③ Desktop display

Choose whether to add the memory bandwidth bar. This bar shows values only after you set up HWiNFO separately. If you are not sure, leave it off and click [Next]. You can change this at any time later.



④ Ready to Install

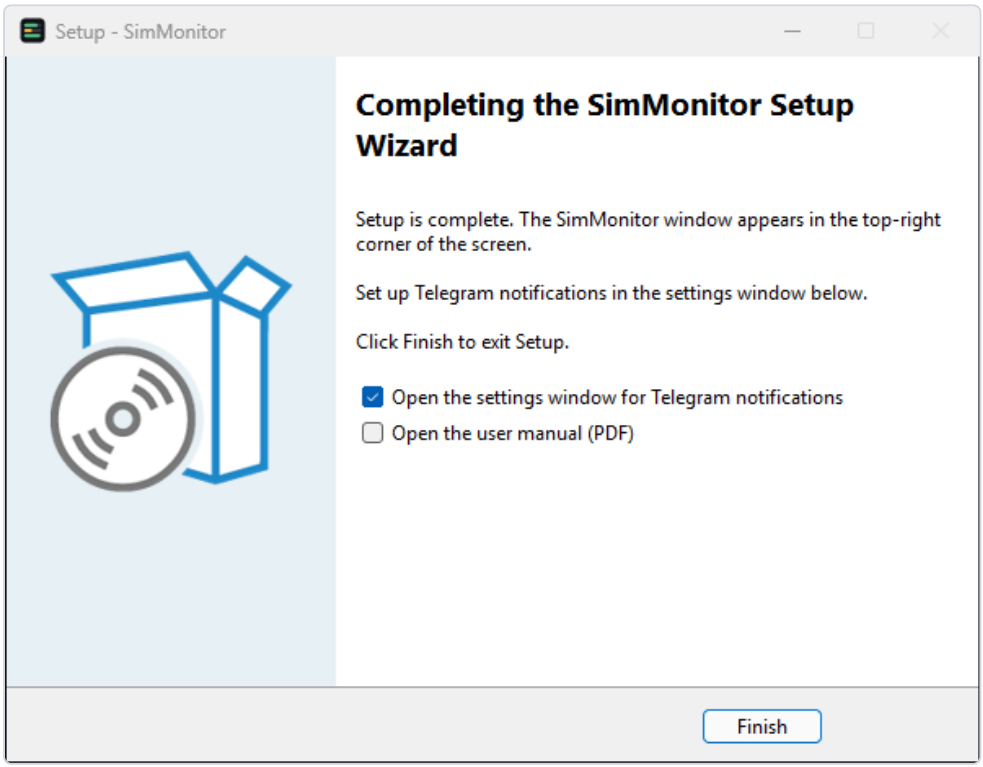
Check what will be installed and click [Install].



If Rainmeter is not installed, setup downloads it from its official source (GitHub) and installs it at this step. If Windows asks **"Do you want to allow this app to make changes to your device?"**, click [Yes].

⑤ Setup complete

If you click [Finish] with [Open the settings window for Telegram notifications] checked, the settings window from section 4 opens right away.



3-4. After setup

The SimMonitor window appears in the top-right corner of the screen. If no simulation is running, it shows "No simulations running".

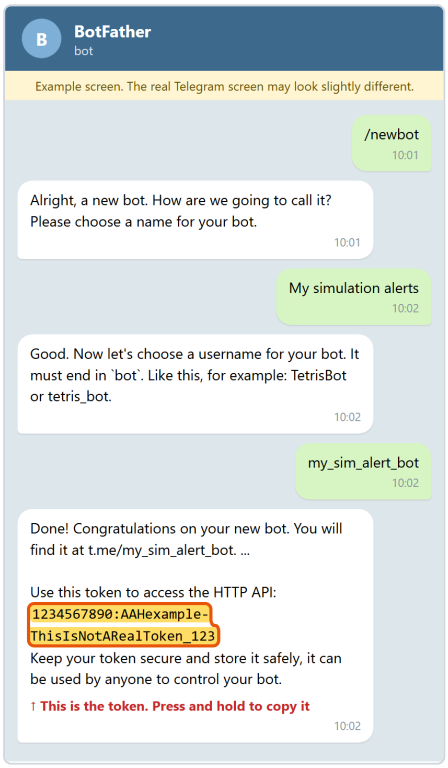
- **Moving the window:** Drag the window with the mouse to where you want it. It remembers the new position.
- **Automatic start:** SimMonitor starts automatically when Windows starts. You do not need to start it yourself.
- **Rainmeter's own windows:** If Rainmeter was installed for the first time, Rainmeter's default windows (illustro), such as a clock and disk usage, may also appear. If you do not need them, right-click that window and choose [Unload skin].

4. Setting up Telegram notifications (optional)

To get notifications on Telegram, create **your own notification bot** and connect it to SimMonitor. This takes about 5 minutes. If you do not need notifications, skip this section.

4-1. Create a bot with BotFather

1. In Telegram, search for **BotFather** and open it. Choose the official account with a blue verification mark next to its name. You can also click [Open BotFather] in the settings window.
2. Send `/newbot`.
3. Send a name for the bot. Any name is fine. Example: `My simulation alerts`
4. Send a username for the bot. Use only English letters, numbers and underscores (`_`), and end it with `bot`. Example: `my_sim_alert_bot`. If the name is already taken, send a different one.
5. In the message from BotFather, press and hold the **token** (a long string of numbers and letters with a colon) to copy it.



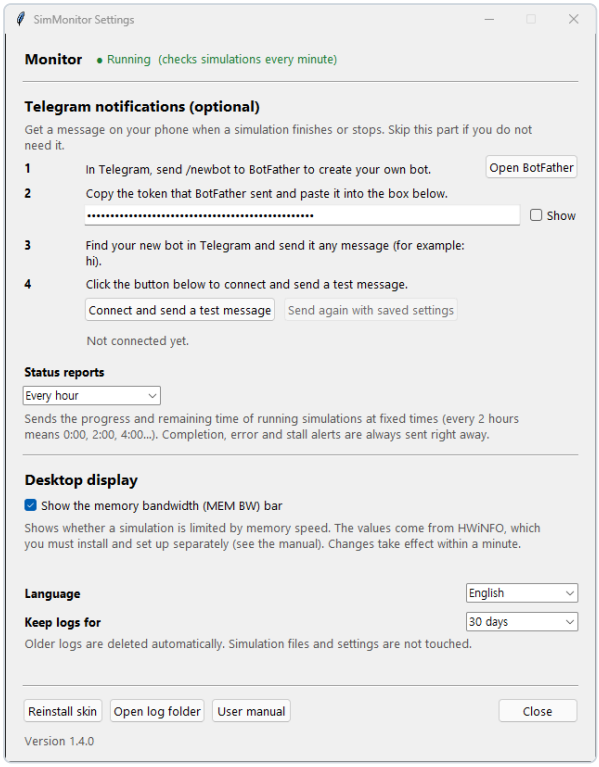
> **The token is like a password.** Do not send it to anyone or post it on the internet. If it was revealed by mistake, send `/revoke` to BotFather to disable that token right away, then connect again with a new token.

4-2. Send a message to your bot

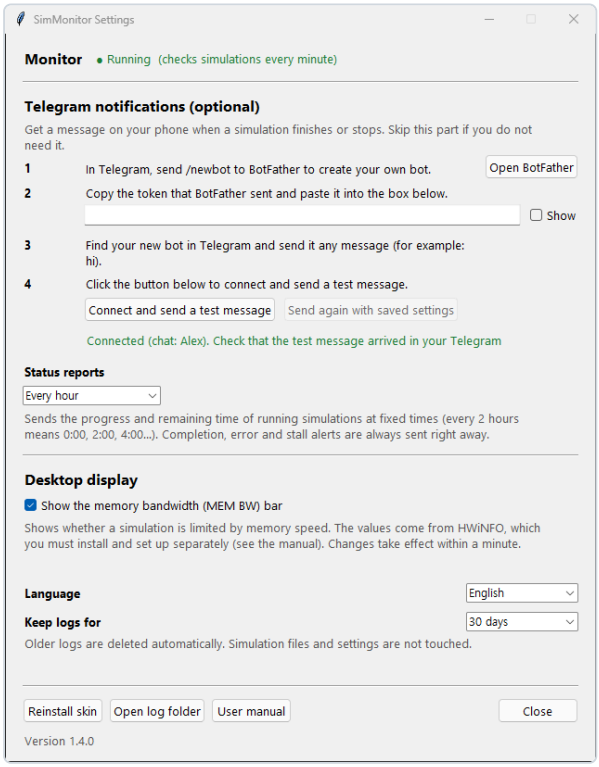
Tap the `t.me/...` link in BotFather's message to open your bot. Tap [Start] or send any message once (for example: `hi`). Your bot can send you messages only after you do this.

4-3. Connect to SimMonitor

1. Open the settings window. Use the window opened at the end of setup, or click **SimMonitor Settings** on the Start menu.
2. Paste the token into the box in step 2 (Ctrl+V). The box hides it with dots.
3. Click **[Connect and send a test message]**.



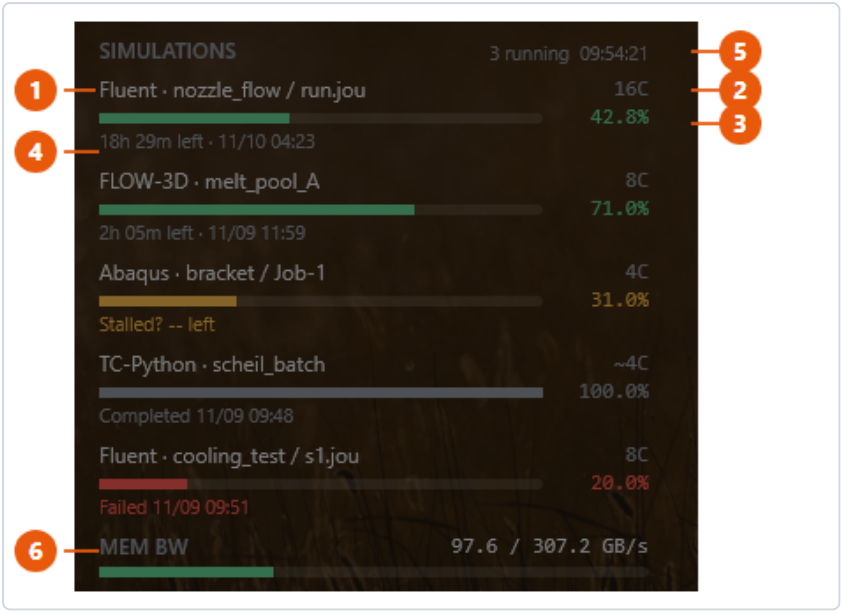
You are done when "Connected (chat: name)" appears in green and a test message arrives in your Telegram. If the name in the brackets is not your Telegram name, someone else sent a message to the bot first. In that case, get a new token with `/revoke` in BotFather and connect again from the beginning.



If red text appears, follow its instructions.

Message	What to do
The bot has not received any messages yet	You skipped 4-2. Send a message to your bot once, then click again
The token is not valid	Check that you copied the whole token exactly
Invalid TELEGRAM_BOT_TOKEN format	Other characters got mixed in before, after or inside the token. Copy and paste it again
Could not connect to Telegram	Check your internet connection. A company firewall may also be blocking Telegram
Choose the chat that should receive notifications, then click again	The bot is in more than one place (a private chat, a group). Choose from the list where you want to receive notifications

5. Reading the display



No.	Meaning
1	Program · simulation name / input file (journal or job name). The simulation name is the name of the simulation folder
2	Number of CPU cores this simulation uses. A tilde, as in `~4C`, means the number is estimated from CPU usage
3	Progress. The bar and the number show the same value
4	Time remaining · expected finish time (month/day hour:minute). This is an estimate based on the speed over the last 30 minutes
5	Number of running simulations · time of the last check. Updates every minute
6	Memory bandwidth: amount in use now / theoretical maximum. Shown only if you turned it on

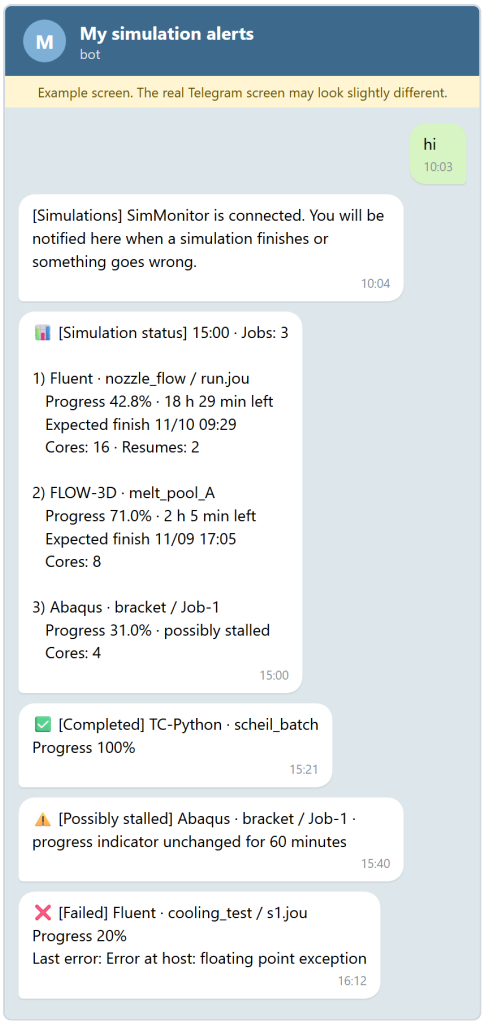
Bar colors

Color	Meaning
Green	Running normally
Yellow	Needs checking. No progress for more than 60 minutes ("Stalled?"), or waiting to be restarted
Red	Ended with an error (Failed, Ended early, Stopped by reboot)
Gray	Finished (Completed). Finished simulations disappear from the list after 15 minutes

Other indicators

- **If the header turns red and shows "Data not updating",** the monitor has not been able to update the information for more than 3 minutes. See section 11.
- **If the progress shows `--`,** SimMonitor could not find this simulation's target (end time or number of steps) in its files. There is nothing wrong with the simulation.
- If there are 9 or more simulations, the header shows `+N more` and only 8 are listed.

6. Types of Telegram notifications

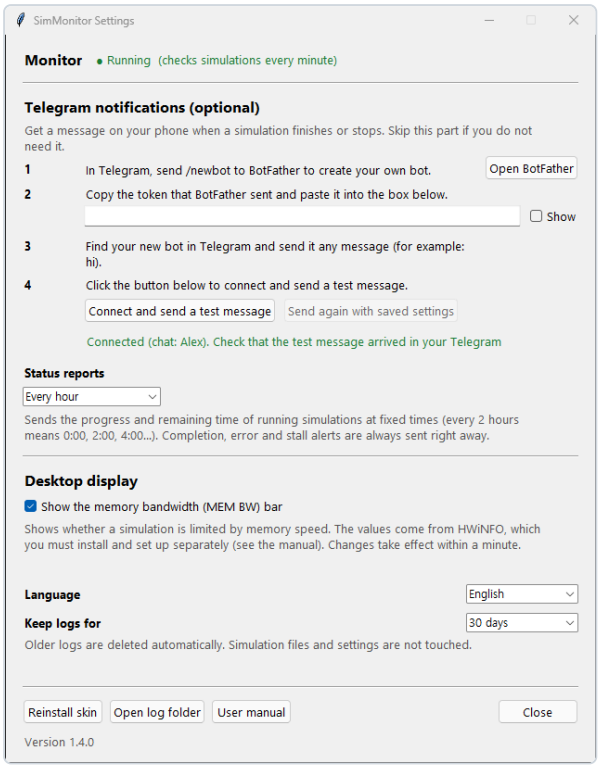


Notification	When it arrives
📊 `[Simulation status] 15:00`	At a set interval (default: every hour, on the hour). Not sent if no simulation is running. You can change the interval as described in section 7
✅ `[Completed]`	When a simulation runs to the end normally
❌ `[Failed]`, `[Ended early]`	When a simulation stopped with an error or ended before reaching its target. The last error line is included
❌ `[Stopped by reboot]`	When the PC restarted during a simulation
⚠️ `[Possibly stalled]`	When the program is still running but has made no progress for 60 minutes (Fluent, FLOW-3D, OpenFOAM)
⚠️ `[Not resumed]`	When the automatic restart script has not restarted the simulation for more than 15 minutes
ℹ️ `[Ended, result unknown]`	When a simulation ended but there is not enough information to tell whether it succeeded

Completion, error and stall notifications are sent as soon as they happen, whatever the interval setting. If the internet is down for a while, unsent notifications are kept (the latest 50) and sent later.

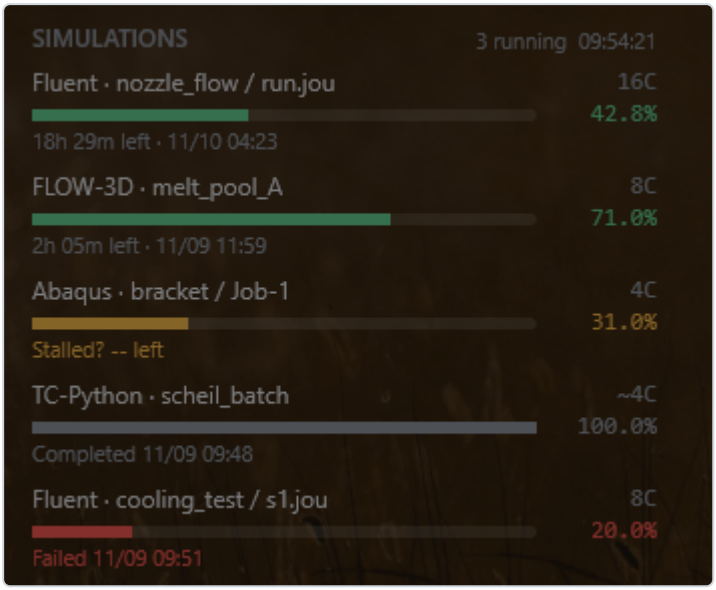
7. Changing settings

Open **SimMonitor Settings** from the Start menu.



What you want to do	How
Start the monitor when it is stopped	[Start] next to "Stopped" at the top
Switch to another bot	Paste the new token and click [Connect and send a test message]
Check that notifications arrive	[Send again with saved settings]
Change the status report interval	In "Status reports", choose Off, Every 30 minutes, Every hour, or Every 2, 3, 6 or 12 hours. Reports are sent at the same times each day (every 2 hours means 0:00, 2:00, 4:00...)
Turn the memory bandwidth bar on or off	The checkbox under "Desktop display". The change takes effect within a minute
Change the language	Choose in "Language" (한국어, English, 日本語, 简体中文, 繁體中文). The settings window switches to the selected language immediately. The desktop window and Telegram notifications switch within a minute
Change how long logs are kept	In "Keep logs for", choose 7, 14, 30, 90 or 180 days (default: 30 days). Older logs are deleted automatically, and the change takes effect within a minute. Simulation files and settings are never deleted
The desktop window has disappeared	[Reinstall skin]
View the log	`simmon-YYYY-MM-DD.log` in [Open log folder]. There is one file per day, and the file with the latest date has the most recent entries

When you turn off the memory bandwidth bar, the bottom part of the window is removed and the window becomes shorter.



8. Using the memory bandwidth bar (optional)

Use this to see whether a memory-heavy simulation is limited by memory speed. The values come from **HWiNFO**, a hardware information program, so you must install HWiNFO and set it up as shown below. If you do not, the bar shows `--`.

> The free version of HWiNFO64 may be used for non-commercial purposes only. If this is a company or work PC, check the HWiNFO license first.

1. Download HWiNFO64 from the official HWiNFO website, install it and start it.
2. In the first window, turn on **Sensors-only** and click [Start].
3. Click the gear icon (Settings) at the bottom of the sensor window and go to the **HWiNFO Gadget** tab.
4. Turn on **Enable reporting to Gadget**.
5. Select **DRAM Read Bandwidth** in the list and turn on **Report value in Gadget**. Do the same for **DRAM Write Bandwidth**.
6. Click [OK]. Leave HWiNFO running. It is convenient to set HWiNFO to start with Windows in its settings.
7. In the SimMonitor settings window, turn on "Show the memory bandwidth (MEM BW) bar".

- **Do not rename the items.** SimMonitor finds the values only if their names (Label) in the Gadget settings stay exactly `DRAM Read Bandwidth` and `DRAM Write Bandwidth`. If a name is different or the unit is not GB/s, SimMonitor shows `--` instead of a wrong value.
- **Some PCs do not have this sensor.** If `DRAM Read Bandwidth` is not in the HWiNFO list, you cannot use this bar on that PC.
- **The maximum on the right is a theoretical value.** It is calculated as number of memory channels in use x speed (MT/s) x 8 bytes. The value you can actually reach is lower.
- **Bar colors:** green below 70% of the maximum, yellow from 70%, red from 85%.

9. TC-Python progress display (optional)

SimMonitor cannot see the progress of a TC-Python (Thermo-Calc) script from the outside. If you do nothing, it shows only whether the script is running and how long it has been running. If you add a few lines to the script, the progress and time remaining are shown too. The calculation works the same even if you do not add them.

```
import os, sys
sys.path.insert(0, os.path.expandvars(r"%LOCALAPPDATA%\Programs\SimMonitor\app"))
import simmon_progress

for i, case in enumerate(cases):
    ...  # one calculation
    simmon_progress.update(i + 1, len(cases))  # number finished, total number
```

`simmon_progress` does not raise an error even if it fails to write its record, so it never stops your calculation.

10. Monitoring OpenFOAM

SimMonitor supports OpenFOAM builds that run directly on Windows. It has been tested with the ESI Windows build; blueCFD-Core is expected to work but has not been tested yet. Both serial runs and parallel runs started with `mpiexec` are shown. OpenFOAM running inside WSL2 or Docker is not supported yet (WSL2 support is planned).

Progress and completion are most accurate when the solver output is saved to a log file. Run the solver like this, or with `runApplication` or `runParallel` in an `Allrun` script:

```
simpleFoam > log.simpleFoam
mpiexec -n 4 simpleFoam -parallel > log.simpleFoam
```

Item	What is shown
Name	Case folder name / solver name (for example `pitzDaily / simpleFoam`)
Cores	Number of processes of a parallel run. A serial run shows 1
Progress	Last `Time =` value in the log / `endTime` in `system/controlDict`, counted from `startTime`
Completed	The log ends with `End`. A run that stops before `endTime` because a steady-state solution converged or a `runTimeControl` condition was met also counts as completed
Failed	`FOAM FATAL ERROR` appeared (the error line is included), or the run stopped without `End`
Ended early	The run stopped before `endTime`, for example with `stopAt writeNow`

- **Without a log file** (output only in the console window), only rough progress from the result time folders (`0.1`, `0.2` ...) is shown. When the run ends, SimMonitor cannot tell whether it succeeded and sends `[Ended, result unknown]`.
- **Name the log file** so that it starts with `log` or ends with `.log` (for example `log.simpleFoam` or `run.log`).
- **If you continue a run with `startFrom latestTime`,** it counts as the same simulation and the number of resumes goes up. If you delete the result time folders and start over, it counts as a new simulation.
- If `endTime` is set with a calculation such as `#calc`, the progress shows `--`, but monitoring and the completion notification still work.

11. Troubleshooting

The SimMonitor window does not appear on the desktop

1. Check whether the Rainmeter icon is among the hidden icons (^) at the bottom right of the taskbar. If it is not there, start Rainmeter from the Start menu.
2. Click [Reinstall skin] in the SimMonitor settings window.

The header is red and shows "Data not updating"

The monitor program is not running. Click [Start] at the top of the SimMonitor settings window. If it stops often, check the `simmon-YYYY-MM-DD.log` with the latest date in [Open log folder].

My simulation is not in the list

- **Check that the program is supported.** Only Fluent, FLOW-3D, Abaqus, Ansys Mechanical, OpenFOAM and TC-Python are supported. For Mechanical, only simulations solved directly on this PC are shown. Simulations sent to a remote solver (RSM) are not shown. For OpenFOAM, only Windows builds are shown. OpenFOAM running inside WSL2 or Docker is not shown yet.
- **Check that it runs on the same PC.** SimMonitor only sees simulations on the PC where it is installed.
- **Wait a moment.** A simulation that has just started appears within a minute.

OpenFOAM progress is only rough, or a finished run is reported as "Ended, result unknown"

The solver output is not being saved to a log file. Save it to a log file, for example `simpleFoam > log.simpleFoam` (section 10). OpenFOAM running inside WSL2 or Docker is not shown yet.

Telegram notifications do not arrive

1. Click [Send again with saved settings] in the settings window.
2. If even the test message does not arrive, check that you have not blocked the bot in Telegram and that the internet works.
3. If it still does not work, go through section 4 again from the beginning.

The installer does not run at all (Smart App Control)

If **Smart App Control** in Windows 11 is on, it blocks unsigned programs. This feature does not allow exceptions for individual programs. You can see its status in [Windows Security] > [App & browser control] > [Smart App Control settings]. On some Windows versions, once you turn it off you cannot turn it back on without reinstalling Windows, so decide carefully. If someone manages security on this PC, ask them first.

12. Updating and uninstalling

Updating: Download the new installer and run it. Your Telegram settings and your desktop display choices are kept. If the SimMonitor settings window is open, setup asks you to close it. Close the window and continue.

Uninstalling:

1. Find SimMonitor in [Settings] > [Apps] > [Installed apps] (Windows 10: [Settings] > [Apps] > [Apps & features]) and click [Uninstall], or click **Uninstall SimMonitor** on the Start menu.
2. During uninstallation, you are asked whether to also delete the Telegram settings and the history. If you plan to install again, click [No].

Rainmeter, which was installed together with SimMonitor, is not removed. If you do not need it, uninstall Rainmeter separately in the same way.

13. Privacy and security

Item	Details
Install location	`%LOCALAPPDATA%\Programs\SimMonitor` (inside your user folder)
Token and settings	The `%LOCALAPPDATA%\sim-monitor` folder. Normally only your Windows account (and the PC's administrators) can open it. The token is stored as plain text in `.env.secrets` in this folder, so take care if several people share the PC
What is sent over the internet	Only Telegram messages (simulation name, progress, time remaining, error line). They go to the Telegram server (`api.telegram.org`)
Simulation files	Read only. SimMonitor never creates or changes files in simulation folders
Screen and log	The token is never shown on screen or written to the log
Log retention	Logs stay only on this PC, in the `%LOCALAPPDATA%\sim-monitor` folder, and are never sent anywhere. They are deleted automatically after the period chosen in "Keep logs for" (default: 30 days)

SimMonitor is free for personal and academic use; for company use, please contact us. The full terms are in `app\LICENSE.txt` in the installation folder and on the GitHub page.